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Designing iron carbide embedded isolated boron (B) and nitrogen (N) atoms co-doped porous carbon fibers networks with tiny amount of B-N bonds as high-efficiency oxygen reduction reaction catalysts

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Abstract

In this paper, polyvinylpyrrolidone (PVP) and 4-hydroxybenzeneboronic acid (o-HPBA) are polarized in alkaline solution and then employed as boron (B) and nitrogen (N) sources. $\text{Fe}(\text{AC})_2/\text{o-HPBA}/\text{PVP}/\text{DMF}$ (N, N-Dimethylformamide) slurries are first electrospun into three-dimensional (3D) networks. Due to the electrostatic repulsion between negatively charged $-\text{O}^-$ groups surrounding N and B atoms, N and B atoms can be kept far away from each other. Upon optimizing o-HPBA dosages, the maximum amount of isolated B and N atoms are successfully doped into the 3D porous iron carbide (Fe_3C) embedded B and N co-doped carbon fibers ($\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$) networks with tiny B-N clusters. Electrochemical measure results demonstrate that $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ exhibit much better oxygen reduction reaction (ORR) activity than 20 wt.% Pt/C in alkaline electrolyte. In acidic electrolyte, $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ also displays ORR activity comparable to Pt/C (its onset and half-wave potentials are just 40 and 50 mV more negative than those of Pt/C). Moreover, $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ show better stability and methanol tolerance ability than Pt/C for ORR in both alkaline and acidic solutions. This work opens new avenues for intentionally preventing the malignant combination of B and N atoms and boosting the electrocatalytic activity of B and N co-doped and Fe_3C embedded carbon catalysts.

Keywords: Boron; Nitrogen; Carbon nanofibers; Hierarchically porous architecture; $\text{Fe}_3\text{C}@C$ units; Oxygen reduction reaction

1. Introduction

Severe consumption of traditional fossil fuels poses more and more challenges to environmental pollution, human health, energy security, etc. Thus, exploring substitution of economic and sustainable clean energy for traditional fossil fuels is becoming urgent.[1] Among those sustainable energy conversion and storage devices (metal-air batteries, fuel cells, etc.), oxygen reduction reaction (ORR) occurring at the cathode is a fairly important electrochemical process. Engineering noble metals-based nanocrystals with size, shape, composition, and structure control is an effective approach to boost the sluggish ORR process.[2] Nevertheless, the high cost, low abundance on earth, fuel crossover effect, together with durability issues hinder their practical application in fuel cells.[3] It is relatively critical to develop nonprecious metal-based catalysts (NPM-BCs) with high electrochemical catalytic performances and much better durability to replace noble-metals for the practical application in various energy conversion devices.[4-8]

Previous advances in NPM-BCs engineering inspire us that electrochemically catalytic performances of NPM-BCs are mainly governed by the intrinsic nature of active sites, amount of accessible active sites, and mass/electron transport ability.[9-13] Consequently, recent significant technological advances for designing highly efficient NPM-BCs have focused on the aforementioned aspects.[14-17] Among them, Fe_3C nanoparticles (NPs) functionalized carbonaceous nanomaterials ($\text{Fe}_3\text{C}@\text{C}$) have been proved to be

one of the most promising NPM-BCs with ORR catalytic performances comparable to noble metals in alkaline electrolyte and even close to noble metals in acidic electrolyte.[18] Subsequent research studies suggested that those activated graphitic carbon layers are the main active sites towards ORR because of the unique host-guest electronic interaction between the inner Fe_3C NPs cores and outer graphitic carbon shells.[19, 20] More recently, doping nitrogen (N) atoms into outer carbon layers has been further proved to boost ORR catalytic performances of $\text{Fe}_3\text{C}@\text{C}$. [1] That is because the lone-pair electrons of N dopants violently activate carbon (C) atoms neighboring N atoms by conjugating with π electrons and therefore the π electrons therefore can be much more effective for boosting ORR catalysis.[21] Up to now, the activation of outer graphitic carbon layers of $\text{Fe}_3\text{C}@\text{C}$ is just confined to N doping. However, experimental and theoretical calculation results attest that doping electron-deficient boron (B) atoms into carbon sheets can also break the electroneutrality of sp^2 C and create charged sites favorable for O_2 adsorption.[3, 22] In recent works, B and N co-doped carbonaceous materials have been synthesized and showed significantly improved electrocatalytic activity toward ORR compared with either N-doped or B-doped ones.[1] With these in mind, the intuition suggests that co-doping B and N atoms into outer carbon layers may be another advanced methodology for further boosting ORR electrocatalysis activities of $\text{Fe}_3\text{C}@\text{C}$ based NPM-BCs.

Opportunities and challenges always exist simultaneously. **Table S1** summarizes nearly all as-reported methodologies related to B and N co-doped carbonaceous materials. It is clear that the simultaneously doping of B and N atoms into carbons through a one-step route results in more than 30 % of N atoms bonded with B atoms, forming a large amount of B-N bonds. The experimental and theoretical investigations collectively proved that the bonded case (*i.e.* B-N bonds) cannot break the inertness of the carbons. Thus, the atomic spin and charge densities that determine the catalytic capability of materials for ORR are not rich in B-N bonds doped carbons. The electron transfer in graphitic carbons with abundant B-N bonds is also difficult during ORR. In other words, B-N bonds doped into carbons would lead to the poor ORR catalysis performances.[23, 24]

In this paper, we adopted a new methodology to synthesize the novel three-dimensional (3D) hybrid networks consist of iron carbide (Fe_3C) embedded into B and N co-doped porous carbon fibers (*i.e.*, $\text{Fe}_3\text{C}@\text{BNPCFs}$). Through the carbonization of as-electrospun [4-hydroxybenzeneboronic acid (o-HPBA)/ $\text{Fe}(\text{AC})_2$ / polyvinylpyrrolidone (PVP)] precursor networks [see details in **Scheme 1 and Section 1 (Experimental Section)** in Supporting Information (**SI**)], the resultant catalysts can be successfully synthesized. In fact, after being polarized in alkaline condition, Fourier-transform infrared (FT-IR) spectra indicate that the negatively charged C-O^- groups are widely distributed surrounding N atoms for PVP. Meanwhile, large amounts of negatively charged

B-O⁻ and -O⁻ groups are also uniformly dispersed neighboring B atoms for o-HPBA. Therefore, the N and B atoms existing in PVP and o-HPBA will be kept far away from each other because of the electrostatic repulsive effect between C-O⁻ groups and B-O⁻/-O⁻. Through the pyrolysis processes in nitrogen (N₂) atmosphere at 900 °C, various Fe₃C@BNPCFs hybrid materials were successfully synthesized. After optimizing the o-HPBA dosages, X-ray photoelectron spectroscopy (XPS) analysis results prove that Fe₃C@B_{1.0}NPCFs have fairly high total N content (7.63 at.%) and total B content (4.48 at.%) with the comparatively low B-N contents (less than 8.2 % of total B or total N atoms). On the other hand, the typical 3D net-like structures of o-HPBA/Fe(AC)₂/PVP precursor networks can be woven by the electrospinning technology. After being calcined in N₂, the resultant Fe₃C@B_{1.0}NPCFs exhibit the 3D net-like channels with abundant hierarchical pores dispersed along each PCF as well.

By utilizing a simple and controllable one-step route based on the electrostatic repulsion interaction between B and N sources, isolated B and N atoms can be homogeneously co-doped into the outer carbon shells of 3D porous Fe₃C@B_{1.0}NPCFs. Being ascribed to the aforementioned advantages refer to content and structure, Fe₃C@B_{1.0}NPCFs exhibit much more excellent ORR catalytic activity than the state-of-the-art Pt/C in alkaline electrolyte. In addition, its ORR catalysis activity measured in acidic electrolyte is also comparable to Pt/C, which is nearly the best result reported up to now. At last,

$\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ even displays better stability and methanol tolerance toward ORR than Pt/C in both alkaline and acidic solutions. This work opens new avenues for intentionally preventing the malignant combination of B and N atoms as well as boosting the electrocatalytic activity of carbon-based catalysts and then exploring the excellent noble metal-free electrocatalysts.

2. Experimental

2.1. Synthesis of 3D $\text{Fe}(\text{AC})_2/\text{o-HPBA}/\text{PVP}$ precursor networks.

All chemicals utilized in this paper were analytical purity and obtained from Aladdin Reagent Co. Ltd., Shanghai, China, they were used as received without further purification. As shown in **Scheme 1**, 1.6 g PVP powder and stoichiometric o-HPBA were dissolved into 20.0 mL N, N-Dimethylformamide (DMF) to produce an o-HPBA-PVP-DMF solution. Then, the as-prepared o-HPBA-PVP-DMF solution was adjusted to alkaline (pH=11) by adding NaOH particles and further stirred for 24 h (for realizing the polarization treatment of o-HPBA and PVP molecules). Then, 0.7 g ferric acetate $[\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}]$ into the as-polarized o-HPBA-PVP-DMF solution and the obtained mixture was further stirred for another 12 h to prepare the homogeneous precursor [*i.e.*, o-HPBA- $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2$ -PVP-DMF] slurry. The electrospinning technique was adopted for weaving o-HPBA- $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2$ -PVP-DMF precursor slurries into $\text{Fe}(\text{AC})_2/\text{o-HPBA}/\text{PVP}$ precursor networks (Details of the conditions for electrospinning technique are displayed in the **Experimental Section 1.1** in the *SI*).

2.2. Synthesis of 3D porous $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ networks

In order to synthesize $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ networks, those as-electrospun o-HPBA/ $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2/\text{PVP}$ precursor networks with different contents were then collected and calcined at 900 °C in N_2 for 2 h. As the comparison, $\text{PVP}/\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2$ precursor networks without adding o-HPBA were also prepared and calcined under the same condition to prepare $\text{Fe}_3\text{C}@B_0\text{NPCFs}$ control sample. In this paper, neither strong acid nor strong alkali solution was used for etching the resultant materials. Herein, for $\text{Fe}_3\text{C}@B_x\text{NPCFs}$, the x value represents the molar ratio of B atoms to Fe atoms [*i.e.* o-HPBA: $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$] in the precursor solutions. For example, $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ represents the molar ratio of o-HPBA: $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$ is 1.0. In order to synthesize different $\text{Fe}_3\text{C}@B_x\text{NPCFs}$, the x values were controlled at 0 (0.70 g $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$), 0.25 (0.70 g $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$ and 97.5 mg o-HPBA), 0.5 (0.70 g $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$ and 194.9 mg o-HPBA), 1 (0.70 g $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$ and 389.8 mg o-HPBA) and 1.5 (0.70 g $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$ and 584.7 mg o-HPBA), by only changing the dosages of o-HPBA. $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot 4\text{H}_2\text{O}$ dosage utilized in each precursor solution is a constant.

2.3. Physical characterization of o-HPBA/ $\text{Fe}(\text{C}_2\text{H}_3\text{O}_2)_2/\text{PVP}$ and

$\text{Fe}_3\text{C}@B_x\text{NPCFs}$ networks.

For the physical characterization methods of resultant samples, one can see details in **Experimental Section 1.2** in *SI*. It's important to note that for

calculating error bars of B and N contents measured by XPS, 3 repeat experiments were adopted.

2.4. Electrochemical characterization of various $Fe_3C@B_xNPCFs$ toward ORR in both acidic and alkaline conditions.

For the electrochemistry characterization methods of as-synthesized catalysts, one can see details in *Experimental Section 1.3* in SI. What needs illustration is that for calculating error bars of ORR catalysis potentials and electron transfer number measured in electrochemical tests, 6 repeat experiments were adopted.

3. Results and discussion

3.1. Exploring the polarization and assembly modes of nitrogen and boron atoms existing in precursors in alkaline condition

Under the existing technology for synthesizing isolated B and N co-doped carbonaceous architectures with less electrochemically inactive B-N bonds, the two-step doping route have been widely developed.[23, 25] As shown in **Table S1**, most previous methods with respect to the two-step doping route mainly adopted boric acid and NH_3 as B and N sources respectively. Even though the B and N sources are same and the operating procedures similar, some papers reported abundant B-N contents,[26, 27] whereas others show correspondingly lower B-N contents.[28, 29] These results demonstrate that the existing two-step doping routes are still uncontrollable by external force. In addition, the two-step doping route is relatively

complex and not favorable for mass-production of isolated B and N co-doped carbons. Taken together, it's still challenging but urgent for designing Fe₃C@isolated B and N co-doped carbon shells just by a simple one-step route to achieve the goal of realizing approximate ORR performances or surpassing precious-metals in acidic condition.

In order to prevent the formation of ORR inactive B-N bonds, two operations were adopted: at first, o-HPBA and PVP were polarized in alkaline condition. FT-IR characterization results of o-HPBA and PVP before and after the polarization process (**Figure S1a**) proved that after polarizing PVP in alkaline condition, most $\text{-CH}_2\text{-}$ chemical bonds successfully transformed into the $\text{R}_2\text{-HC-O-H}$ [30] and the N-C=O changed into the N=C-O-H groups by the typical enol interconvert reaction.[31] Especially, the peak of C-O-H groups becomes wider and shows a remarkable red shift because the C-O-H is mild acid and C-O^- groups shows stronger polarity, showing that those as-appeared C-O-H groups have thoroughly changed into the C-O^- groups in alkaline solution. For the o-HPBA (**Figure S1b**), after the polarization processes, the peak of B-O-H bonds almost disappeared but a new peak of borate group (B-O^-) appeared. Meanwhile, the peak of -O-H groups conjugated with benzene rings became wider and also displayed a remarkable red shift. This phenomenon is due to that the -O-H groups conjugated with benzene rings are weakly acidic and the -O^- groups display stronger polarity compared with -O-H groups. These results also prove that -O-H groups are

changed into the O^- groups. Thus, there is a trend that the B atoms in o-HPBA molecules would keep a distance from the N atoms in PVP molecules because of the electrostatic repulsion between $\text{B-O}^-/\text{O}^-$ groups surrounding B atoms in o-HPBA and O^- groups surrounding N atoms in PVP (**Figure S2**). On the other hand, the as-assembled o-HPBA/ $\text{Fe}(\text{AC})_2$ /PVP precursor slurries were weaved into 3D o-HPBA/ $\text{Fe}(\text{AC})_2$ /PVP by the simple electrospinning technology, which can prevent B in o-HPBA and N in PVP touching with each other due to their layer upon layer stacking.

3.2. Structural characterization of precursor networks and as-pyrolyzed catalysts

Before the calcination process, scanning electron microscopy (SEM) images reveal the typical 3D net-like architectures of all o-HPBA/ $\text{Fe}(\text{AC})_2$ /PVP precursor networks (**Figure S3**), demonstrating the successful structure design for precursors. After the carbonization process at 900 °C in N_2 atmosphere, the SEM (**Figures 1a1-e1**), high-resolution SEM (**Figures 1a2-e2**), and transmission electron microscopy (TEM) images (**Figures 1a3-e3**) reveal that the 3D net-like architectures of precursor networks were perfectly kept for resultant $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ ($x=0, 0.25, 0.5$, and 1.0) with rough surfaces. Nevertheless, after the pyrolysis process, the long precursor nanofibers were cracked into smallish carbonaceous rods for $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$ (**Figures 1e1-e3**). Meanwhile, the vast NPs appeared for all resultant samples exhibiting the largest difference in morphologies compared with precursor networks. The diameter distribution histograms of PCFs (**Figures 1a4-e4**) and size distribution histograms of the appeared NPs (**Figure S4**) have also been

recorded. The average diameters of PCFs for all control materials increase by the order of $\text{Fe}_3\text{C@NPCFs}$ (76.61 nm) < $\text{Fe}_3\text{C@B}_{0.25}\text{NPCFs}$ (93.44 nm) < $\text{Fe}_3\text{C@B}_{0.5}\text{NPCFs}$ (99.96 nm) < $\text{Fe}_3\text{C@B}_{1.0}\text{NPCFs}$ (111.30 nm) < $\text{Fe}_3\text{C@B}_{1.5}\text{NPCFs}$ (134.43 nm). Nevertheless, the average sizes of metal NPs embedded into carbon sheets decrease by an opposite order of $\text{Fe}_3\text{C@NPCFs}$ (29.62 nm) > $\text{Fe}_3\text{C@B}_{0.25}\text{NPCFs}$ (26.48 nm) > $\text{Fe}_3\text{C@B}_{0.5}\text{NPCFs}$ (24.71 nm) > $\text{Fe}_3\text{C@B}_{1.0}\text{NPCFs}$ (22.31 nm) > $\text{Fe}_3\text{C@B}_{1.5}\text{NPCFs}$ (21.36 nm). One can see that o-HPBA can increase the diameters of PCFs and simultaneously limit the sizes of metal NPs, mainly because the o-HPBA molecules can afford the extra carbon sources. The smaller NPs sizes can further increase the interaction between metal NPs (cores) and outer carbon layers (shells) and then afford more active centers.[32, 33] On the other hand, due to the abundant O elements of o-HPBA, the H_2O and CO_2 gases were released in the pyrolysis process of precursors.[34, 35] Thus, when excess amounts of o-HPBA were utilized for $\text{Fe}_3\text{C@B}_{1.5}\text{NPCFs}$, the superabundant gas accessory substances led to the collapse of 3D networks. In view of the structural integrity, $x=1.0$ is the best choice for preparing the ideal $\text{Fe}_3\text{C@BNPCFs}$ with large amounts of tiny NPs@C units dispersed along the surfaces.

In a catalytic system, the porous characteristics directly control the absolute amount of exposed active centres and mass transfer efficiency of catalysts. This means that the significance of porous structures in boosting catalytic activity is equal to that of active sites.[36] High-resolution TEM (HRTEM) image (**Figure 2a**) shows that various mesopores dispersed along

surfaces of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$, which is ascribed to the pyrolyzation of oxygen-containing functional groups and the release of H_2O and CO_2 gases during the carbonation process. [34, 35] These mesopores can afford better exposure and enhance the utilization of active sites on catalyst surfaces.[37, 38] HRTEM image of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ also indicates that metal NPs are tightly enclosed by graphitic carbon layers (with the lattice spacing of 0.36 nm) (**Figure 2b**). The well-resolved lattice fringes with the interval distance of 0.21 nm are ascribed to the (211) planes of Fe_3C (**Figure 2c**). On the other hand, the crystal plane angle of the (-211) and (211) planes is $\sim 68.4^\circ$, which is also attributed to Fe_3C phase.[19] The crystalline structures of different $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ were further characterized by typical X-ray diffraction (XRD) patterns (**Figure 2d**). For all samples, the wide diffraction peaks located between 20° and 30° suggest the formation of graphitic carbon (which can promote the electron transmission) during the pyrolysis. The characteristic peaks located at 37.8° , 43.9° , 45.0° , 46.0° , 49.2° , and 54.5° are corresponding to the Fe_3C phase (JCPDS, no. 89-2867).[39] Because of the severe shrinkage of $\text{Fe}_3\text{C}@B_{0.25}\text{NPCFs}$ (**Figures 1a1-a4**) and $\text{Fe}_3\text{C}@B_{0.5}\text{NPCFs}$ (**Figures 1b1-b4**), some Fe_3C NPs were exposed on surfaces and oxidized into Fe_2O_3 (JCPDS, no. 87-0722).[40] However, the diffraction peaks of Fe_2O_3 are fairly weak in XRD pattern of $\text{Fe}_3\text{C}@B_{0.5}\text{NPCFs}$. Finally, the diffraction peaks of Fe_2O_3 cannot be found in XRD patterns of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$. Both

electron microscopy and XRD characterization results prove the successful building of 3D porous $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ networks.

The porosities of different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ were determined by N_2 adsorption-desorption isotherms (**Figure 2e**). As shown in **Table S2**, $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ exhibit the largest BET surface area ($158.3 \text{ m}^2 \text{ g}^{-1}$) compared with $\text{Fe}_3\text{C}@\text{PCFs}$ ($91.9 \text{ m}^2 \text{ g}^{-1}$), $\text{Fe}_3\text{C}@\text{B}_{0.25}\text{NPCFs}$ ($134.3 \text{ m}^2 \text{ g}^{-1}$), $\text{Fe}_3\text{C}@\text{B}_{0.5}\text{NPCFs}$ ($143.5 \text{ m}^2 \text{ g}^{-1}$), and $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ ($125.1 \text{ m}^2 \text{ g}^{-1}$). The corresponding pore size distribution plots (**Figure 2f**) suggest the formation of abundant mesopores distributed between 2 and 40 nm for each sample. On the other hand, $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ display the maximum amount of mesopores located between 4-20 nm. Raman spectra were collected to assess the graphitic structures and defects within the frameworks of resultant materials (**Figure 2g**). Both graphitic D and G bands can be perfectly seen at about 1364 and 1580 cm^{-1} for all control materials. $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ exhibit the largest I_D/I_G ratio of 1.51 compared with other reference samples (**Table S2**), indicating that the maximum amount of graphitic defects and edge sites have been sculptured along surfaces of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ after the pyrolysis process. It is clear that $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ display obvious structural advantages compared with all other reference samples.

3.3. Surface element content of as-pyrolyzed catalysts

Elemental distribution statuses of different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ samples were investigated by energy-dispersive x-ray diffraction (EDX) method (**Figure 3**).

It is clear that C (**Figures 3a2-e2**) and O (**Figures 3a3-e3**) elements are all uniformly distributed along the surfaces of $\text{Fe}_3\text{C}@B_x\text{NPCFs}$. However, the strength of Fe signal decreases by an order of $\text{Fe}_3\text{C}@B_{0.25}\text{NPCFs} > \text{Fe}_3\text{C}@B_{0.5}\text{NPCFs} > \text{Fe}_3\text{C}@B_{1.0}\text{NPCFs} > \text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$ (**Figures 3a4-e4**). As most of the Fe_3C NPs are concentrated inside the $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$, the Fe signals tested by EDX are fairly weak. The tenacious Fe signals tested by EDX are mainly derived from Fe atoms doped on surfaces of PCFs in the form of Fe-N-C bonds. As shown in **Figure 3a5**, no B atoms can be found in elemental mapping of $\text{Fe}_3\text{C}@B_{0.25}\text{NPCFs}$. Nevertheless, with rising dosages of o-HPBA in precursor slurries (**Figures 3b5-e5**), the B signals increase gradually by an order of $\text{Fe}_3\text{C}@B_{0.25}\text{NPCFs} < \text{Fe}_3\text{C}@B_{0.5}\text{NPCFs} < \text{Fe}_3\text{C}@B_{1.0}\text{NPCFs} < \text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$. The N contents increase gradually by the order of $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs} < \text{Fe}_3\text{C}@B_{1.0}\text{NPCFs} < \text{Fe}_3\text{C}@B_{0.5}\text{NPCFs} < \text{Fe}_3\text{C}@B_{0.25}\text{NPCFs}$ (**Figures 3a6-e6**) and the $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$ exhibit a tiny N content decrease.

XPS was further performed to characterize the elemental configurations of different dopants. As shown in **Figure 2h**, XPS survey spectra of different $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ indicate that o-HPBA can decrease the Fe contents on surfaces of PCFs (**Table S3**). The Fe content of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ is 0.21 at.%, once again proving that Fe elements mainly concentrate in inside the PCFs. This result is consistent with those of XRD (**Figure 2d**) and EDX characterization (**Figure 3**). The B contents of $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ increase

gradually along with raising x values. Especially, $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$ display the maximum B content of 7.08 at.%, which is larger than that of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ (4.49 at.%). However, $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ possesses the maximum N content of 7.63 at.%, which is much larger than that of $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$ (4.49 at.%). EDX and XPS measurement results all indicate that the dosages of o-HPBA can affect the final N contents doped into carbon sheets. On the one hand, the C atoms in o-HPBA act as carbon sources as well, which can increase the diameters of PCFs; on the other hand, the abundant oxygen-containing functional groups existed in o-HPBA will release large amounts of gas bubbles in the pyrolysis processes and then create vast porous structures along the surfaces of PCFs. Due to the structural advantages, more accessible N atoms have been doped into PCFs. Finally, one can see that the heavily doping B and N atoms into carbon layers indeed leads to the decrease of O based groups exposed along surfaces of PCFs.

FT-IR spectra were further tested to probe the nature of N and B atoms dispersed into the carbonaceous skeletons of $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ networks. As shown in **Figure 2i**, $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ do not show any B signals and the strength of N-C bonds is comparatively weak. After the B-doping, three extra adsorption peaks for B-N, B-C, and N-C bonds can be observed for $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ as well in addition to C=C, C-O-C [the ether C-O-C linkages (crosslinking, dehydroxylation)], and O-H bonds. By increasing the dosages of o-HPBA, B-N content of each sample changes weakly and only weak B-N signals can be

observed in FT-IR spectra of $\text{Fe}_3\text{C}@B_x\text{NPCFs}$. This result directly proves that our B and N co-doping technology can really prevent the building of B-N bonds, sharply raising the doping amounts of isolated B and N atoms. On the other hand, the B-C contents rise gradually from $\text{Fe}_3\text{C}@B_{0.25}\text{NPCFs-900}$ to $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs-900}$. The N-C contents first rise to the maximum value for $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and then decrease slightly for $\text{Fe}_3\text{C}@B_{1.5}\text{NPCFs}$. These characterization results are in good agreement with those of EDX measurements.

High-resolution N 1s and B 1s XPS spectra were further fitted to reveal the bonding configurations of $\text{Fe}_3\text{C}@B_x\text{NPCFs}$. Both N 1s and B 1s spectra for all control samples display distinct characteristic peaks, indicating the totally different chemical statuses of B and N dopants. One can see that except for $\text{Fe}_3\text{C}@B\text{NPCFs}$ (in which B-N bonds cannot be found), the high-resolution N1s spectra of other $\text{Fe}_3\text{C}@B_x\text{NPCFs}$ samples can be fitted into six parts assigned to B-N (396.7 ± 0.2 eV), pyridinic N ($\sim 398 \pm 0.2$ eV), pyridinic N-Fe (399.3 ± 0.3 eV), pyrrolic N (400.5 ± 0.1 eV), graphitic N (401.7 ± 0.1 eV), and oxidized N (403.6 ± 0.3 eV) species, respectively (**Figures 4a-e**). [41-43] It is important to note that only the pyridinic-N, M-(pyridinic) N-C, and graphitic N species can act as active sites toward ORR catalysis [41, 44]. Seeing from the comparison of the results referring to N 1s recorded in **Figure 5a and Table S4**, the total content of ORR active graphitic N, pyridinic-N, and Fe-(pyridinic)N-C species for $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ (5.163 at.%) is much larger than those of other

control samples. $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ display a small B-N content (0.62 at.%; ~8.09 % of total N content), which is similar to other control samples. Furthermore, B1s spectra of resultant $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ can be fitted into five parts corresponding to B_4C (less than 188 eV), BC_3 (189.6-190 eV), B-N (191 eV), BC_2O (191.8 eV), and B-O (192.6 eV) (**Figures 4a1-e1**). [45, 46] As shown in **Figure 5b and Table S4**, the contents of ORR active BC_3 species [47] reach the maximum value of 2.21 at.% for $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$. Although $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ has a larger total B content, its BC_3 content (2.22 at.%) is almost similar to that of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (2.21 at.%). Even though there were changes in dosages of o-HPBA in the precursor slurries, the B-N contents calculated from N 1s and B 1s XPS spectra are all less than 8.2 % of total N and B atoms in $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$. The combination of the structural features such as large diameters of PCFs, maximum amount of porous structures, largest BET surface area and most carbon edges of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ have provided necessary environments for doping more N and B based species.

To prove the efficiency of the electrostatic repulsion interaction between precursor molecules in preventing the formation of ORR inactive B-N species, the high resolution N 1s (**Figure 5c**) and B 1s spectra (**Figure 5d**) of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ synthesized without polarization of the precursors were also recorded. As shown in **Figures 5e, f**, compared to $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ with the polarization of precursors, $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ synthesized without polarization of precursors have less pyridinic N (including Fe-(pyridinic)N-C), graphitic N,

and BC₃ ORR active centres accompanied by more ORR inactive B–N bonds. These results indicate that the electrostatic repulsion between –O[–] surrounding the B and N atoms indeed insures the separation of B and N atoms and then impedes the construction of vast ORR inactive B–N bonds in the resultant materials. Therefore, our technology of co-doping abundant isolated B and N atoms into the outer carbon sheets of Fe₃C@C by a simple one-step method is feasible and highly efficient.

3.4. ORR catalytic activity test in alkaline electrolyte solution (0.1 M KOH)

The ORR electrocatalytic activity and stability of Fe₃C@B_xNPCFs catalysts were performed by rotating disk electrode (RDE) and rotating ring-disk electrode (RRDE) techniques in both 0.1 M KOH and 0.5 M H₂SO₄ electrolytes. ORR polarization linear sweep voltammetry (LSV) curves of Fe₃C@B_xNPCFs and 20 wt.% Pt/C were first tested in O₂-saturated 0.10 M KOH (**Figure 6a**). The ORR catalytic activity was measured for Fe₃C@B_{1.0}NPCFs in terms of onset potential (E_{onset}), half-wave potential ($E_{1/2}$), and diffusion-limiting current (j_L , recorded at 0.6 V vs. RHE). Compared to other control catalysts (**Figure 6b** and **Table S5**), Fe₃C@B_{1.0}NPCFs exhibit the largest E_{onset} , $E_{1/2}$, and j_L values. Compared with the state-of-the-art 20 wt.% Pt/C catalyst, the optimal Fe₃C@B_{1.0}NPCFs still exhibit much more excellent catalytic kinetic parameters toward ORR: E_{onset} (0.968 versus 0.954 V vs. RHE), $E_{1/2}$ (0.832 versus 0.813 V vs. RHE), and j_L (4.83 versus 4.71 mA cm^{–2}). Moreover, these parameters also outperform most previously reported noble metal-free ORR catalysts (as shown in **Table S5**). All these compared results

demonstrate the most excellent ORR catalysis efficiency of Fe₃C@B_{1.0}NPCFs.

Fe₃C@B_xNPCFs were further evaluated for their kinetic performances by using the Tafel plots (**Figure 6c**). Tafel slope got on Fe₃C@B_{1.0}NPCFs is just 60.8 mV decade⁻¹, which is much smaller than those of other reference catalysts (including Pt/C: 64.5 mV decade⁻¹). This result reveals that transfer of first electron on Fe₃C@B_{1.0}NPCFs is the rate determining step under Temkin conditions for the adsorption of intermediates [48]. The electron transfer numbers (*n*) per O₂ molecule for ORR have been calculated from LSV curves in RDE tests (**Figure S5**) according to the Koutechy-Levich (*K-L*) plots (**Figure S6**) by using the equations expressed as eqn. (S1) and eqn. (S2) in SI. The as-calculated *n* values are displayed in **Figure 6d**. The average *n* value of Fe₃C@B_{1.0}NPCFs recorded between 0.4-0.7 V vs. RHE (~4.00) is larger than other reference catalysts (including Pt/C), suggesting a four electron pathway of Fe₃C@B_{1.0}NPCFs toward ORR in 0.1 M KOH.

To further evaluate the ORR catalytic performances of Fe₃C@B_{1.0}NPCFs and Pt/C, we conducted the typical RRDE measurements (**Figure 6e**). The *n* values can also be calculated from eqn. (S3) and the H₂O₂ formation yields were calculated from eqn. (S4) based on the as-collected RRDE data. The corresponding measure results indicate that H₂O₂ yield measured on Fe₃C@B_{1.0}NPCFs remains below 3 % in a potential range of 0.25-0.8 V vs. RHE (**Figure 6f**), which is lower than that of Pt/C. This indicates that Fe₃C@B_{1.0}NPCFs own the extremely high ORR catalytic efficiency in 0.1 M KOH. In addition, the average *n* value calculated between 0.25-0.8 V vs. RHE from RRDE measurements is ~4.00 for Fe₃C@B_{1.0}NPCFs, which

is slightly larger than that measured on Pt/C (~ 3.98). This further confirms the desired four-electron oxygen reduction process of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$.

The transfer coefficient (α) and rate constant (k_{orr}) values of ORR in alkaline condition have been calculated using as-measured Tafel slope b and i_0 values in Tafel plots through eqn. (S5) and eqn. (S6) shown in section 1.3 of SI. In 0.1 M KOH, the α values of optimal $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and commercial Pt/C are 0.97 and 0.92 respectively. Furthermore, the k_{orr} values of ORR for optimal $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and commercial Pt/C are 2.42×10^{-3} and $2.08 \times 10^{-3} \text{ g s}^{-1}$, respectively. All these results provide evidence for the excellent ORR catalysis activity of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$, which even surpasses the commercial Pt/C.

In addition to excellent ORR catalytic activity, $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ also display high selectivity of ORR with strong tolerance to CH_3OH crossover effect in 0.1 M KOH. From **Figure 6g**, it is clearly seen that $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ maintain stable current density after adding 1.0 M CH_3OH into electrolyte; whereas, a significant decrease in current density is observed for Pt/C once 1.0 M CH_3OH is added into 0.1 M KOH electrolyte. The 20 wt% Pt/C even shows obvious oxidation current densities of CH_3OH . Also, $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ show much better stability for ORR than Pt/C catalyst in 0.1 M KOH, as revealed by a stability test in **Figure 6h**.

3.5. ORR catalytic activity test in acidic electrolyte solution (0.5 M H_2SO_4)

Up to now, the bland ORR catalytic performances of NPM-BCs in acidic media are the main drawbacks that limit their practical application in fuel cells. As shown in **Figure 7a and Table S6**, ORR catalytic activity of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ is much more

excellent than other control samples in O₂-saturated 0.5 M H₂SO₄ electrolyte in terms of E_{onset} , $E_{1/2}$, and j_L . Specially, the E_{onset} (0.833 V vs. RHE) and $E_{1/2}$ (0.638 V vs. RHE) values of Fe₃C@B_{1.0}NPCFs are 40 and 50 mV more negative than those of Pt/C catalyst. This standard even surpasses those of other reported NPM-BCs ORR catalysts exhibited in **Table S6**. All these compared results demonstrate the excellent ORR catalysis efficiency of Fe₃C@B_{1.0}NPCFs in acidic conditions. Tafel plots of different Fe₃C@B_xNPCFs networks and Pt/C catalyst have also been measured to evaluate their kinetic performances (**Figure 7b**). In acidic electrolyte, Tafel slope of Fe₃C@B_{1.0}NPCFs (71.8 mV decade⁻¹) is still smaller than those of other control samples including Pt/C (72.4 mV decade⁻¹). These results indicate that the transfer of first electron on Fe₃C@B_{1.0}NPCFs is the rate determining step.

The n values were calculated from RDE test (**Figure S7**) according to the K - L plots (**Figure S8**). The calculated n values of Pt/C and different Fe₃C@B_xNPCFs networks are displayed in **Figure 7c** and the average n values are all ~4.00 for Pt/C and Fe₃C@B_{1.0}NPCFs between 0.2-0.4 V vs. RHE. This value is larger than those of other reference catalysts and suggests the four electron pathway toward ORR in 0.5 M H₂SO₄ for Fe₃C@B_{1.0}NPCFs. RRDE polarization curves (**Figure 7d**) were also recorded for testing H₂O₂ yields and n values of Fe₃C@B_{1.0}NPCFs and Pt/C. The H₂O₂ yields calculated on Fe₃C@B_{1.0}NPCFs are all below ~4.5 % in acidic solution, which are a little larger than those of Pt/C catalyst (~3.5 %) over the measured potential range (**Figure 7e**). Furthermore, the average n value of Fe₃C@B_{1.0}NPCFs calculated between 0-0.65 V vs. RHE through RRDE method is ~3.97 (**Figure 7e**),

which is almost similar to that calculated from Pt/C (~ 3.99), confirming the desired four-electron ORR process. All the test results indicate that $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ indeed exhibit the extremely high ORR catalytic efficiency in 0.5 M H_2SO_4 .

The transfer coefficient (α) and rate constant (k_{orr}) values of ORR in acidic condition have also been calculated using as-measured Tafel slope b and i_0 values in Tafel plots through eqn. (S5) and eqn. (S6) shown in section 1.3 of SI. In 0.5 M H_2SO_4 , the α values of optimal $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and commercial Pt/C are 0.817 and 0.824 respectively. Furthermore, the k_{orr} values of ORR for optimal $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and commercial Pt/C are 2.23×10^{-3} and $2.57 \times 10^{-4} \text{ g s}^{-1}$, respectively. All these results demonstrate the excellent ORR catalysis activity of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$, which is a little lower than the commercial Pt/C in acidic condition.

In addition, the polarization curve recorded after 5000 cycles just displays a 20 mV of negative shift for $E_{1/2}$ toward $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$, whereas there is a 140 mV loss in $E_{1/2}$ for Pt/C under the same test conditions (**Figure 7f**). The above results demonstrate the superior ORR catalysis durability of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ compared with Pt/C in acidic condition. Finally, we measured the possible crossover effect of $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ and Pt/C caused by CH_3OH . Obviously, the current densities recorded over the working potential range for $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ remains unchanged in the presence of 1.0 M CH_3OH (**Figure 7f**), while Pt/C shows a significant CH_3OH oxidation current density in the chosen potential range. This reveals that our noble metal-free $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ show high selectivity for ORR with strong tolerance to crossover effect of CH_3OH in 0.5 M H_2SO_4 .

As shown in **Scheme 2**, previous papers have demonstrated that Fe-N, Pyridinic N, Graphitic N, BC₃, BC₂O, Fe₃C@C, and defect sites can act as ORR active sites [20, 41-47] whereas Pyrrolic N, Oxidized N, B-N, B₄C, and B-O bonds cannot act as ORR active sites. [22-24, 41] On the other hand, abundant porous structures can also boost the dispersion of active sites and mass transfer, which further promote the ORR catalysis efficiency. [44, 45] Based on the structural and compositional characterization results, five underlying factors are responsible for the superior ORR activity and stability of Fe₃C@B_{1.0}NPCFs in both acidic and alkaline solutions: (1) the graphitic structure promotes the electron transport in ORR catalysis. (2) The 3D interpenetrated networks of Fe₃C@B_{1.0}NPCFs are favourable to afford plenty of channels for mass transference and then decrease the internal resistance. (3) The abundant mesopores dispersed on surfaces guarantee the largest BET surface area and maximum edge defect sites of Fe₃C@B_{1.0}NPCFs, which are beneficial to expose large amount of active centres along the surfaces and then increase the utilization factor of electrochemically active sites. (4) Fe₃C@B_{1.0}NPCFs display the smallest particle sizes compared with other control samples. These tiny Fe₃C NPs uniformly embedded into BNPCFs further enhance the unique host-guest electronic interaction between inner Fe₃C NPs cores and outer graphitic carbon shells. This electron interaction promotes the ORR catalytic activity of outer graphitic layers because of their outstanding ability to adsorb oxygen. (5) The electrostatic repulsion between

polarized precursors really prevents the formation of ORR inactive B-N bonds (less than 8.2 % of total N or total B atoms). The vast majority of N and B atoms have been separately doped into the 3D networks. These isolated N and B atoms can break the electroneutrality of sp^2 carbon to create charged sites favorable for O_2 adsorption on the surface. Taken together, the excellent ORR electrocatalytic activity, anti-methanol crossover effect, and long-term stability of $Fe_3C@B_{1.0}NPCFs$ make it a promising low-cost and highly efficient non-noble metal electrocatalyst for ORR catalysis especially in acid electrolyte.

4. Conclusions

In summary, after being polarized in alkaline condition, the electrostatic repulsion force between oxygen-containing functional groups surrounding nitrogen (N) and boron (B) atoms in precursor networks will restrict the formation of ORR inactive B-N bonds in calcining process in the optimal catalyst (*i.e.* iron carbide embedded into B and N codoped porous carbon fibers ($Fe_3C@B_{1.0}NPCFs$)). In addition, profiting from the electrostatic spinning technology, the resultant catalyst owns the three-dimensional and hierarchically porous structures wove by $Fe_3C@B_{1.0}NPCFs$ nanofibers. This kind of structure boosts the absolute quantity of accessible active sites (*i.e.* pyridinic N, pyridinic N-Fe, graphitic N and BC_3) along catalyst's surfaces and meanwhile promotes the mass transfer in oxygen reduction reaction (ORR) catalysis. Electrochemical experiment results reveal that the optimal $Fe_3C@B_{1.0}NPCFs$ exhibits much better catalysis activity than commercial Pt/C

toward ORR in 0.10 M KOH. Additionally, ORR onset potential (E_{onset}) and half-wave potential ($E_{1/2}$) values for $\text{Fe}_3\text{C}@B_{1.0}\text{NPCFs}$ are just 40 and 50 mV more negative than those of Pt/C in acidic solution. This work opens new avenues for intentionally preventing the malignant combining of B and N atoms by the form of B-N bonds and boosting the electrocatalytic activity of B and N co-doped catalysts.

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Figure captions

Scheme 1. The synthesis processes of isolated B and N co-doped 3D porous $\text{Fe}_3\text{C}@\text{BNPCFs}$ networks by a simple and tuneable one-step approach.

Figure 1 SEM images (**a1, b1, c1, d1 and e1**), high-resolution SEM (HRSEM) images (**a2, b2, c2, d2 and e2**), HRTEM images (**a3, b3, c3, d3 and e3**), and diameter distribution histograms of PCFs (**a4, b4, c4, d4 and e4**) for different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ networks calcined at 900 °C. Different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$: $\text{Fe}_3\text{C}@\text{NPCFs}$ (**a1, a2, a3 and a4**), $\text{Fe}_3\text{C}@\text{B}_{0.25}\text{NPCFs}$ (**b1, b2, b3 and b4**), $\text{Fe}_3\text{C}@\text{B}_{0.5}\text{NPCFs}$ (**c1, c2, c3 and c4**), $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (**d1, d2, d3 and d4**), and $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ (**e1, e2, e3 and e4**).

Figure 2. (a) TEM image, (b) high-resolution TEM image, and (c) lattice fringes of a NP for $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ networks. (d) XRD patterns, (e) N_2 adsorption-desorption isotherms, (f) pore size distribution curves, (g) Raman spectra, (h) XPS survey spectra, and (i) FTIR spectra of different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ networks ($x=0, 0.25, 0.5, 1.0$, and 1.5).

Figure 3 SEM images (**a1-e1**) and the corresponding elemental mapping of C (**a2-e2**), O (**a3-e3**), Fe (**a4-e4**), B (**a5-e5**), and N (**a6-e6**) elements (seeing from top to bottom respectively) for resultant $\text{Fe}_3\text{C}@\text{NPCFs}$ (**a1-a6**), $\text{Fe}_3\text{C}@\text{B}_{0.25}\text{NPCFs}$ (**b1-b6**), $\text{Fe}_3\text{C}@\text{B}_{0.5}\text{NPCFs}$ (**c1-c6**), $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (**d1-d6**), and $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ (**e1-e6**) samples.

Figure 4. High resolution N 1s XPS spectra of $\text{Fe}_3\text{C}@\text{NPCFs}$ (**a**), $\text{Fe}_3\text{C}@\text{B}_{0.25}\text{NPCFs}$ (**b**), $\text{Fe}_3\text{C}@\text{B}_{0.5}\text{NPCFs}$ (**c**), $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (**d**), and $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ (**e**), respectively. High resolution B 1s XPS spectra of $\text{Fe}_3\text{C}@\text{NPCFs}$ (**a1**), $\text{Fe}_3\text{C}@\text{B}_{0.25}\text{NPCFs}$ (**b1**), $\text{Fe}_3\text{C}@\text{B}_{0.5}\text{NPCFs}$ (**c1**), $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (**d1**), and $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ (**e1**), respectively.

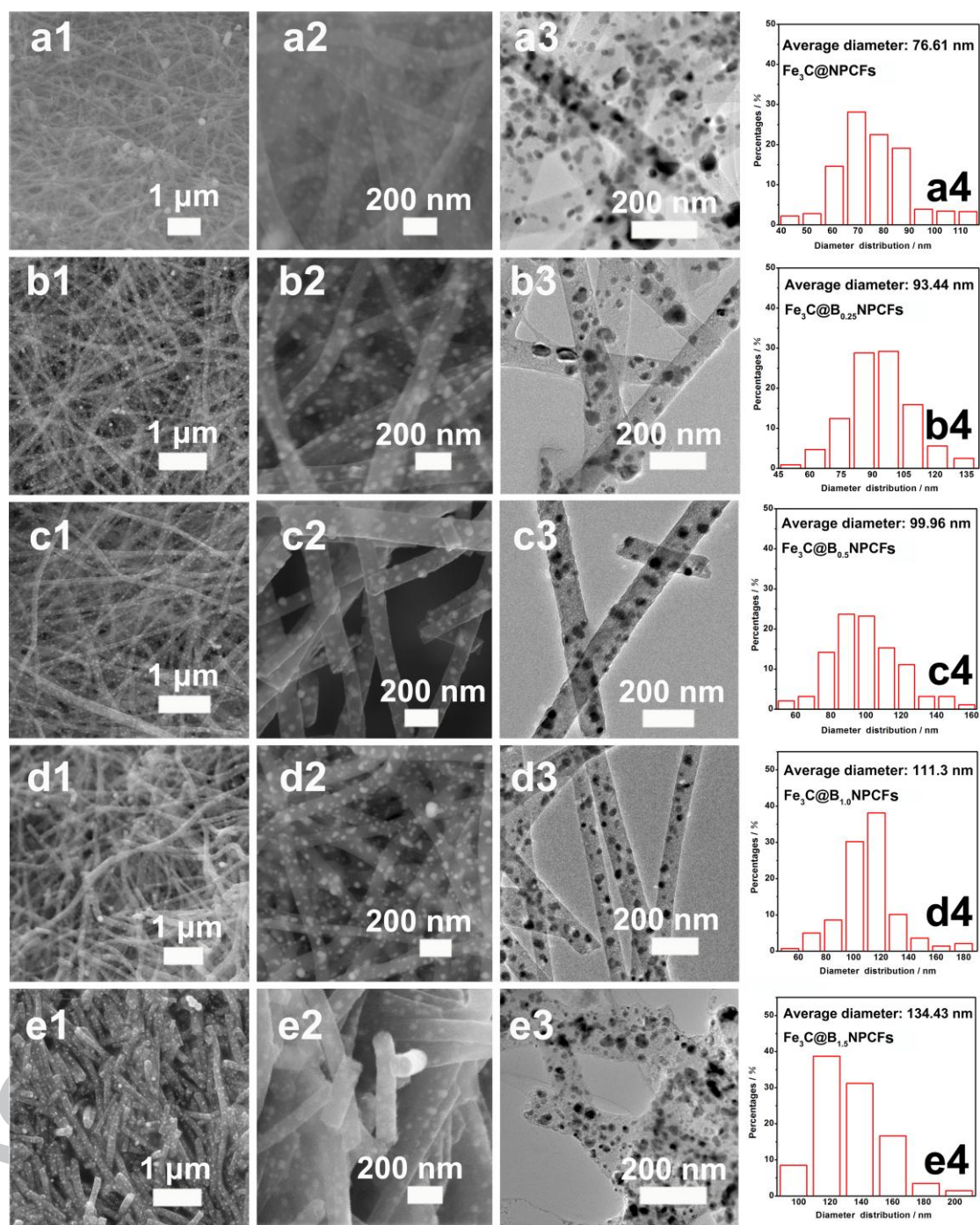
Figure 5 (a) The comparison in contents of B-N, pyridinic N, pyridinic N-Fe, pyrrolic N, graphitic N, and oxidized N revealed in high resolution N 1s XPS plots of resultant $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ ($x=0, 0.25, 0.5, 1.0$ and 1.5) networks. (b) The comparison in contents of B_4C , BC_3 , B-N, BC_2O , and B-O bonds revealed in high resolution B 1s XPS plots of resultant $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ ($x=0, 0.25, 0.5, 1.0$ and 1.5) networks. High resolution N 1s (c) and B 1s (d) XPS spectra of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ synthesized without the polarization of precursors. The comparison in content percentages of N (e) and B (f) species for $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ synthesized with (red) and without (black)

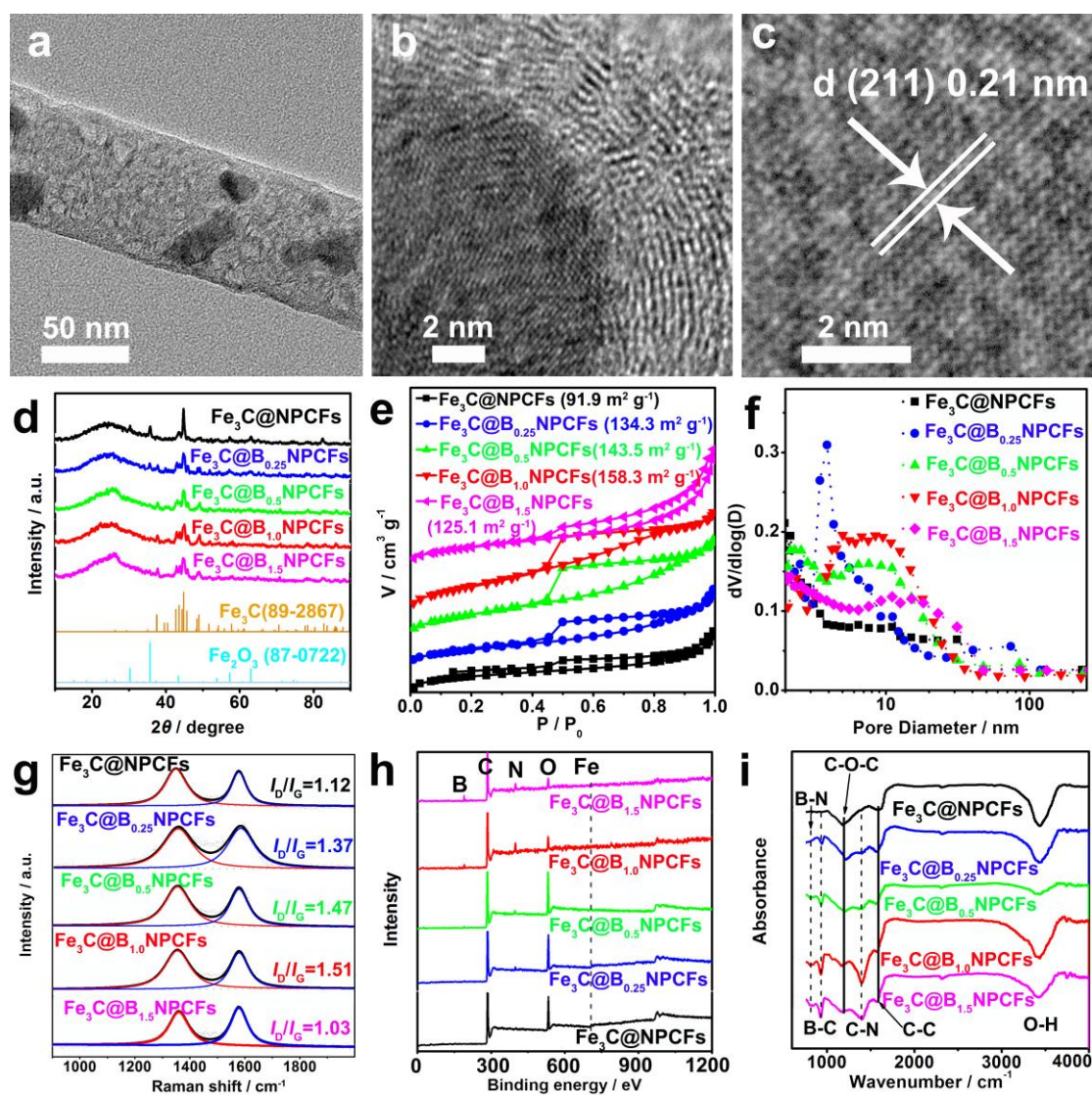
the polarization of precursors in alkaline condition.

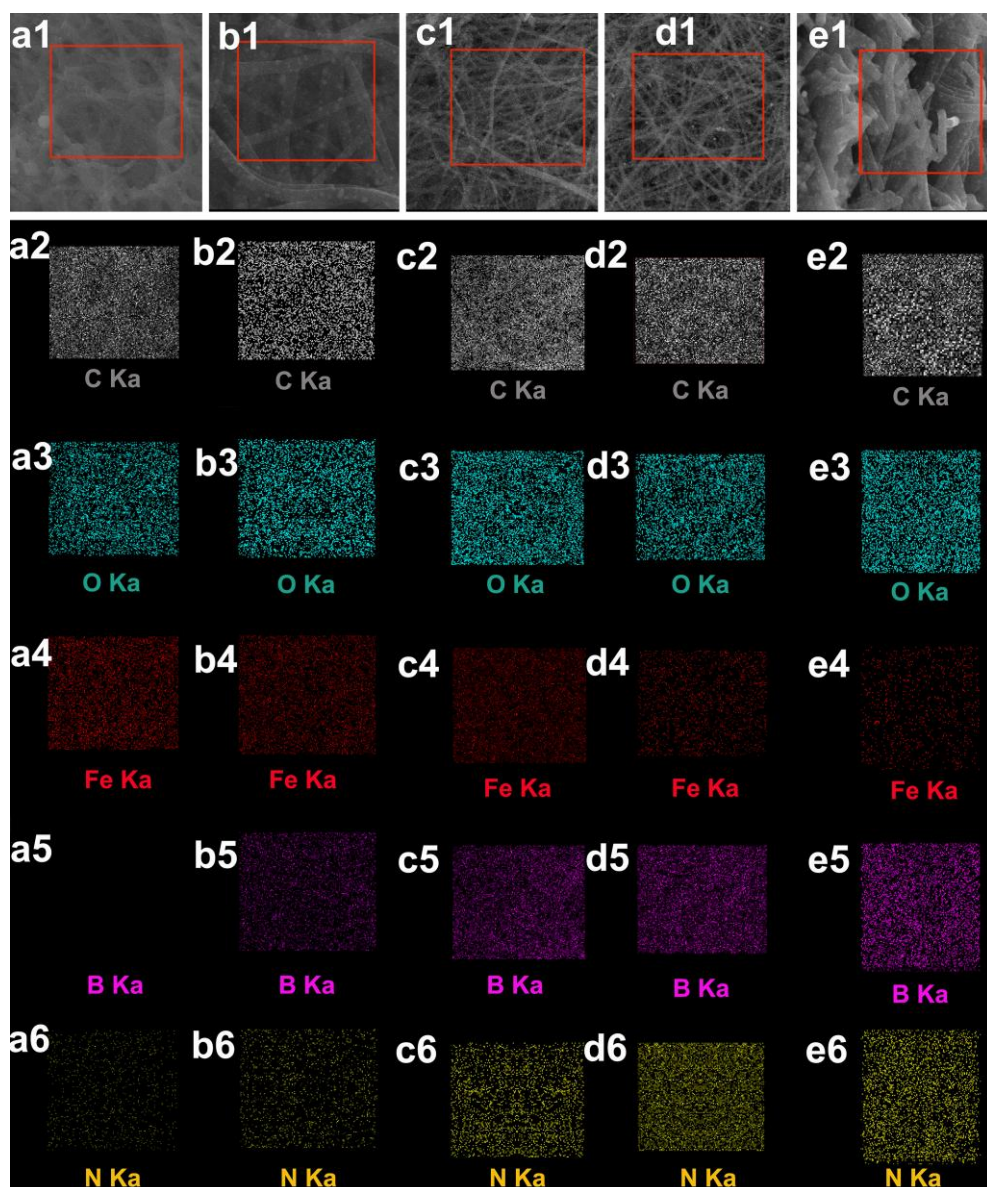
Figure 6 (a) ORR polarization LSV curves, (b) ORR E_{onset} and $E_{1/2}$ values, (c) Tafel plots (Materials in **Figure c**: $\text{Fe}_3\text{C}@\text{NPCFs}$ (c1), $\text{Fe}_3\text{C}@\text{B}_{0.25}\text{NPCFs}$ (c2), $\text{Fe}_3\text{C}@\text{B}_{0.5}\text{NPCFs}$ (c3), $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (c4), $\text{Fe}_3\text{C}@\text{B}_{1.5}\text{NPCFs}$ (c5), and Pt/C (c6) networks), and (d) electron transfer number (n) values calculated from RDE tests for different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ ($x=0, 0.25, 0.5, 1.0$ and 1.5) networks and Pt/C catalyst in O_2 -saturated 0.1 M KOH . (e) I_D and I_R values tested by RRDE method and (f) H_2O_2 yields (solid lines) and n values (dot lines) calculated from RRDE tests toward ORR on $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (red lines) and $20 \text{ wt\% Pt/C}$ (black lines) in O_2 -saturated 0.1 M KOH electrolyte. (g) LSV curves of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (red lines) and Pt/C (black lines) in O_2 -saturated 0.1 M KOH with (dot lines) and without (solid lines) $1.0 \text{ M CH}_3\text{OH}$. (h) LSV curves of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (red lines) and Pt/C (black lines) in O_2 -saturated 0.1 M KOH before (solid lines) and after 5000 cycles (dot lines). Scan rate: 5 mV s^{-1} . Rotation rate: 1600 rpm .

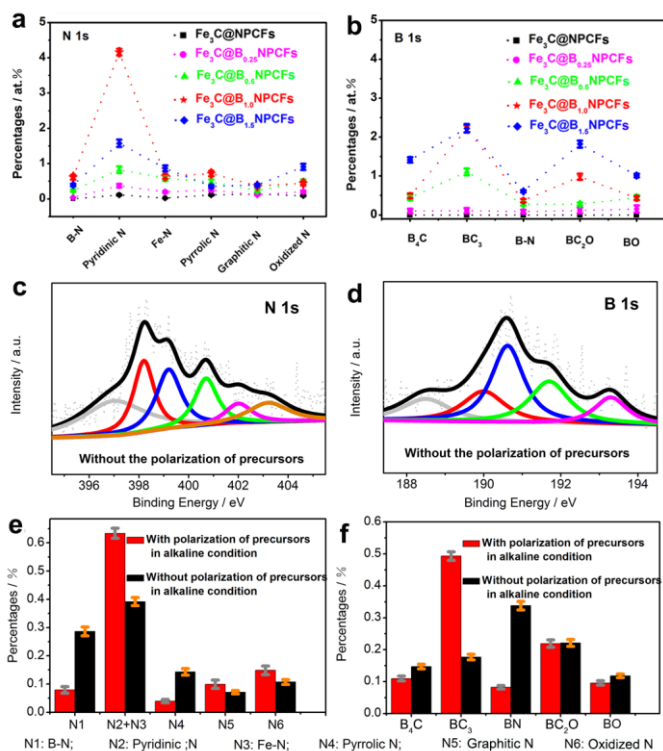
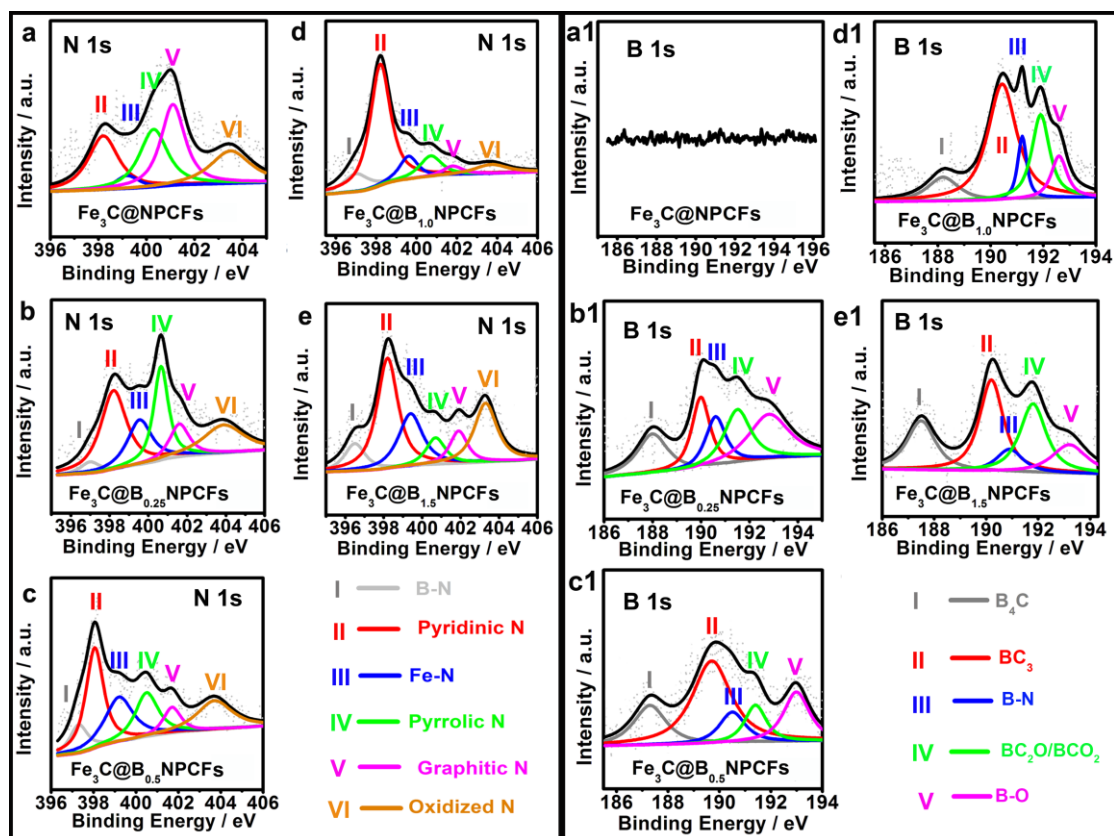
Figure 7 (a) ORR polarization LSV curves, (b) Tafel plots, and (c) n values calculated from RDE tests for different $\text{Fe}_3\text{C}@\text{B}_x\text{NPCFs}$ ($x=0, 0.25, 0.5, 1.0$ and 1.5) networks and Pt/C catalyst in O_2 -saturated $0.5 \text{ M H}_2\text{SO}_4$. (d) I_D and I_R values tested by RRDE method and (e) H_2O_2 yields (solid lines) and n values (dot lines) calculated from RRDE tests toward ORR on $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (red lines) and $20 \text{ wt\% Pt/C}$ (black lines) in O_2 -saturated $0.5 \text{ M H}_2\text{SO}_4$ electrolyte. (f) LSV curves of $\text{Fe}_3\text{C}@\text{B}_{1.0}\text{NPCFs}$ (solid lines) and Pt/C (dot lines) recorded in O_2 -saturated $0.5 \text{ M H}_2\text{SO}_4$ with (green lines) and without (black lines) $1.0 \text{ M CH}_3\text{OH}$; before (black lines) and after 5000 cycles (red lines). Scan rate: 5 mV s^{-1} . Rotation rate: 1600 rpm .

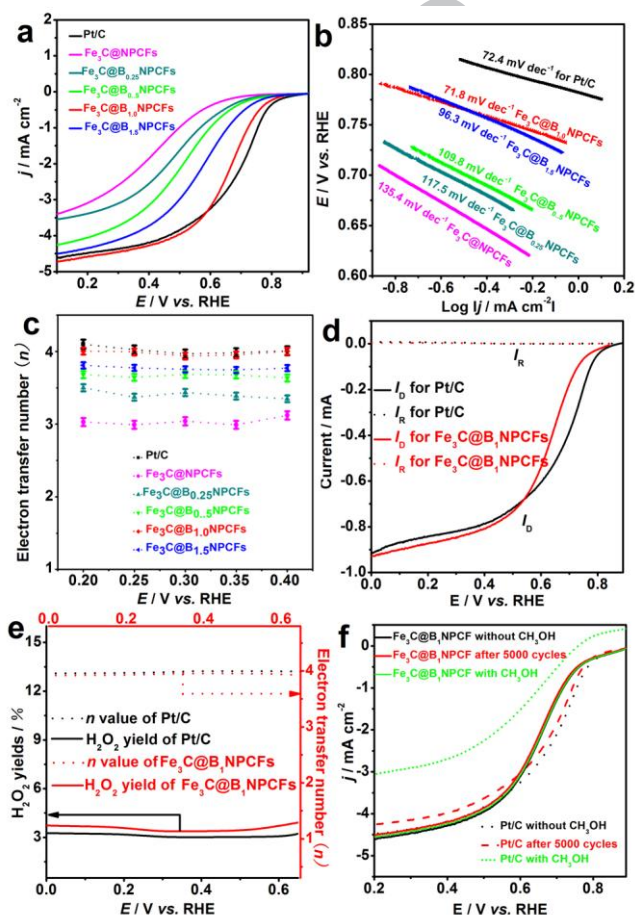
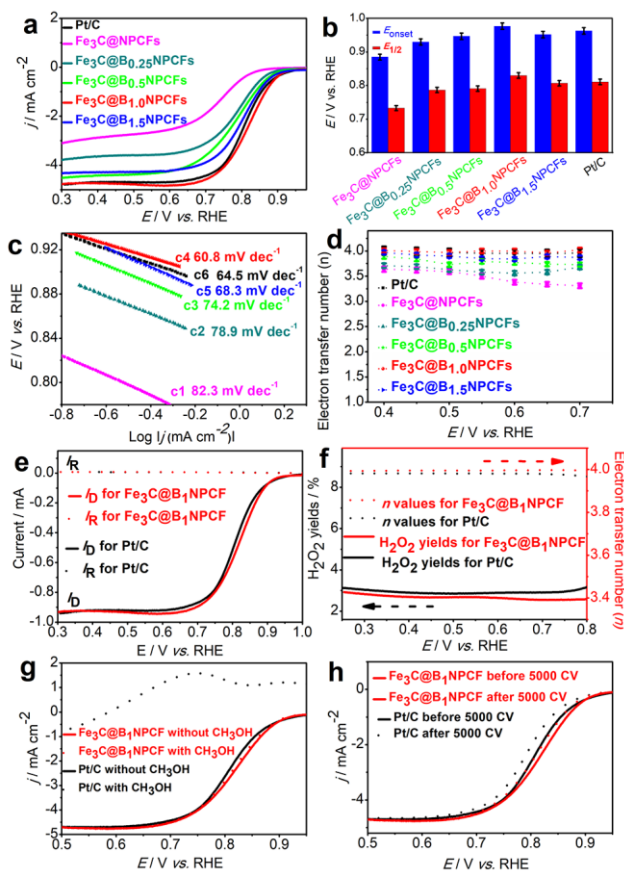
Scheme 2. The key factors affecting the ORR catalysis activity of B and N co-doped 3D porous $\text{Fe}_3\text{C}@\text{BNPCFs}$ networks.

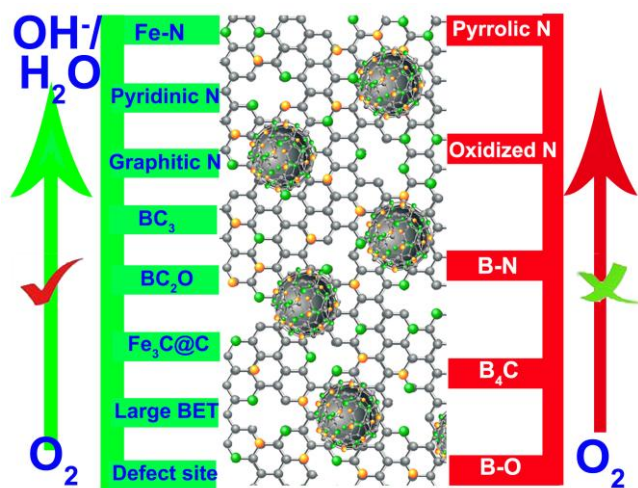
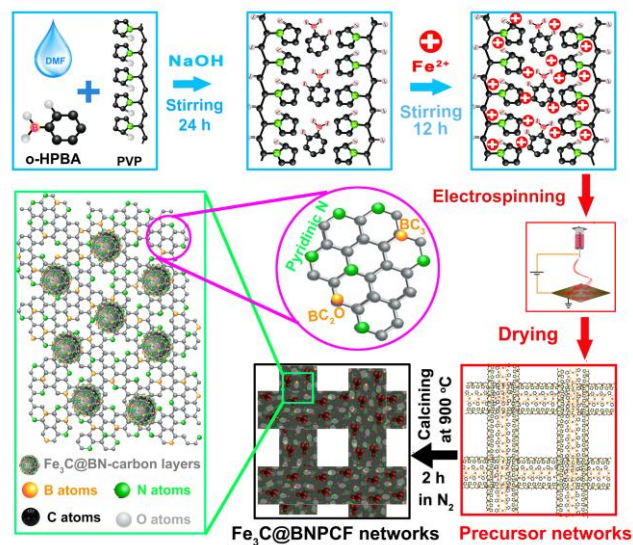




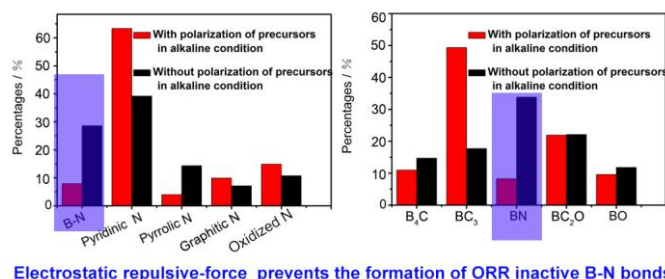








Graphical Abstract



The B-N bond contents calculated from either N or B XPS spectra are less than 8.2 % of total N or B atoms for 3D porous Fe₃C@B_{1.0}NPCFs-900 networks as the electrostatic repulsion effect between precursors.